

## BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

# Atomic Structure

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

## A Retrieval

1. State the relative charge and relative mass of a proton, a neutron and an electron.
2. Define atomic number and mass number.
3. What is the same, and what is different, in a pair of isotopes?

## B Knowledge

4. A neutral atom of  $^{39}\text{K}$  has atomic number 19. Calculate the numbers of protons, neutrons and electrons.
5. Explain why atoms are electrically neutral.
6. Describe where almost all of the mass of an atom is found, and why the atom is mostly empty space.

## C Application and calculation

7. Complete a table for  $^{24}\text{Mg}$ ,  $^{24}\text{Mg}^{2+}$ ,  $^{19}\text{F}$  and  $^{19}\text{F}^{-}$  showing protons, neutrons and electrons.
8. Gallium occurs as 60.0%  $^{69}\text{Ga}$  and 40.0%  $^{71}\text{Ga}$ . Calculate  $A_r$ .
9. Explain why the  $A_r$  of chlorine on the periodic table is not a whole number.

## D Exam-style practice

10. A student claims that  $^{14}\text{C}$  and  $^{14}\text{N}$  are isotopes of each other. Explain why this is incorrect. [2]
11. Calculate  $A_r$  for an element that is 75% isotope-63 and 25% isotope-65. [2]
12. Explain how a 2- ion forms from a neutral atom, referring to protons and electrons. [3]

## E Challenge

13. Neon has peaks at mass numbers 20, 21 and 22 with relative abundances 90.5, 0.27 and 9.25. Calculate  $A_r$  to 3 s.f. and comment on the effect of ignoring the middle isotope.